

Slack Interactive Approval Workflow

Automated Decision Management & Verification Pipeline via n8n

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Platform:	n8n Automation Engine	Integration:	Slack API, Block Kit, Webhooks
Security:	HMAC-SHA256 Verification	Status:	Production Ready / Sanitized Template

1. Executive Summary

This project delivers an enterprise-grade automated approval workflow built on **n8n** and integrated with **Slack**. The system enables team members to initiate request approvals instantly using a custom Slack Slash Command (`/approve-request`). Upon execution, the workflow posts an interactive Slack Block Kit card featuring real-time "Approve" and "Reject" action buttons.

To guarantee enterprise security, every incoming payload is cryptographically authenticated using Slack's **HMAC-SHA256 signature verification** protocol. Once an approving authority interacts with the message, the workflow dynamically updates the original Slack message state, preserving an immutable record of the action taken and the identity of the decision-maker.

2. Key Features & Capabilities

- **Instant Slash Command Trigger:** Custom `/approve-request` trigger allows rapid initiation directly from active Slack channels.
- **Interactive Block Kit UI:** Visually distinct UI elements featuring color-coded action buttons (Green for Approve, Red for Reject).
- **Cryptographic Request Verification:** Custom Node.js JavaScript code nodes compute expected HMAC-SHA256 signatures to prevent unauthorized requests or web-spoofing.
- **Replay Attack Protection:** Automatic rejection of incoming HTTP requests with timestamps older than 5 minutes (300 seconds).
- **Dynamic State Synchronization:** Replaces interactive buttons post-click with an updated status banner including the Slack User ID of the approver/rejecter.
- **Sanitized Template:** Environmentally safe workflow architecture ready for GitHub deployment without exposing secret credentials.

3. Workflow Architecture & Node Matrix

Node Name	Node Type	Functional Purpose
Slash Command Webhook	Webhook (POST)	Listens for incoming /approve-request slash command payloads from Slack.
Verify Slash Signature	Code (JavaScript)	Validates x-slack-signature header against raw payload using HMAC-SHA256.
Respond Processing	Respond to Webhook	Sends immediate HTTP 200 acknowledgment to Slack within the 3-second timeout limit.
Send Approval Message	Slack Node	Posts Block Kit interactive message card into the channel containing action buttons.
Interaction Webhook	Webhook (POST)	Captures button interaction events sent to /slack-interactions endpoint.
Verify Button Signature	Code (JavaScript)	Authenticates button click HTTP request to prevent unauthorized state alteration.
Approve or Reject	Switch Node	Routes execution path based on payload action value (approve vs reject).
Mark Approved / Rejected	Slack Node	Updates original message text, replacing buttons with user-stamped final decision.

4. Security & Cryptographic Verification

The workflow implements robust signature verification following Slack's official security standard:

```
const baseString = 'v0:' + timestamp + ':' + rawBody;
const expected = 'v0=' + crypto.createHmac('sha256', signingSecret)
    .update(baseString)
    .digest('hex');
```

Security Best Practice Compliance

All API Keys and Signing Secrets are externalized using standard placeholder variables (YOUR_SLACK_SIGNING_SECRET), rendering the export file safe for public repositories like GitHub.

5. Deployment & Setup Guide

1. **Import Workflow:** Open n8n → Workflows → Import from File → Select `Slack_Approval_Workflow.json`.
2. **Configure Credentials:** Set up Slack OAuth / Bot Token credentials in n8n.
3. **Add Signing Secret:** In `Verify Slash Signature` and `Verify Button Signature` nodes, enter your Slack Signing Secret in `let signingSecret = '...'`.
4. **Connect Slack App:** Copy Webhook URLs from n8n into Slack App's *Slash Commands* and *Interactive Components* settings.
5. **Publish:** Toggle workflow status to **Active** and execute `/approve-request` in Slack.